UMS logo | PEREKA 2026 logo | Faculty / school logo

PEREKA 2026 · Pertandingan Reka Cipta

Universiti Malaysia Sabah · 13–14 October 2026 · *Transforming Ideas Into Impact*

CLUSTER · EDUCATION | CATEGORY · EAT / ASH | ENTRY ID · _____

AN INTERACTIVE MATHEMATICS LABORATORY

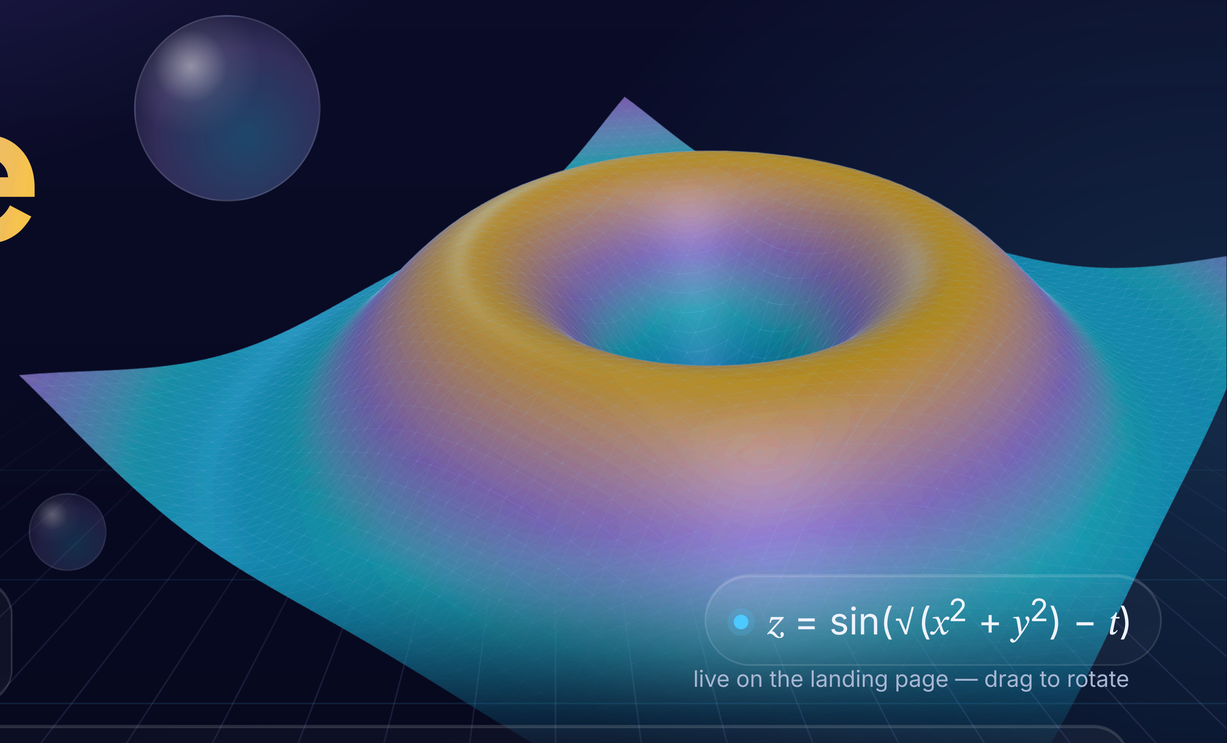
MathXplore

See. Change. Code.

A browser-based lab where students **see** a function, **change** it with a slider, **predict** what happens next — and write the **Python** that draws it. Mathematics and programming, finally taught together.

TEAM Member 1 · Member 2 · Member 3

SUPERVISOR Supervisor name — FROM Faculty / school, institution



5 Explore modules — from lines to 3D and geometry

12 What If? scenarios — predict first, then see

6 levels of Python — taught through maths

12 challenges graded — by running the code

182 automated checks — all passing

0 internet or accounts — needed to run

01 THE PROBLEM

Maths and code, taught apart

Two subjects, no bridge
Loops learnt on arbitrary examples; graphs never linked to code.

Answers without understanding
A calculator answers only what you already knew how to ask.

Errors that teach nothing
`SyntaxError: invalid syntax` is a dead end for a beginner.

02 OUR IDEA

Every Python concept arrives because a piece of mathematics needs it — and every picture on screen is also a program.

MATHEMATICS $y = ax^2 + bx + c$ ⇄ PICTURE ⇄ PYTHON

```
y = a*x**2
  + b*x + c
```

Change any one and the other two follow — a slider rewrites the code, and the code redraws the graph.

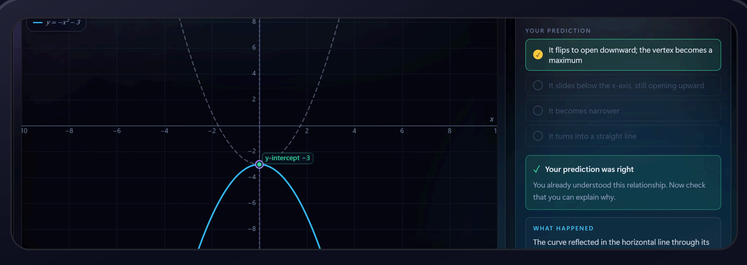
One learning loop, lit up on every working screen

so a student always sees where they are — and that there is a next step

LEARN (6-level pathway) · EXPLORE (5 modules) · PREDICT (What If?) · CHANGE (live sliders) · VISUALISE (2D · 3D) · CODE (CodeLab) · TEST (challenges) · UNDERSTAND (error lessons) · CREATE (sandbox · shapes)

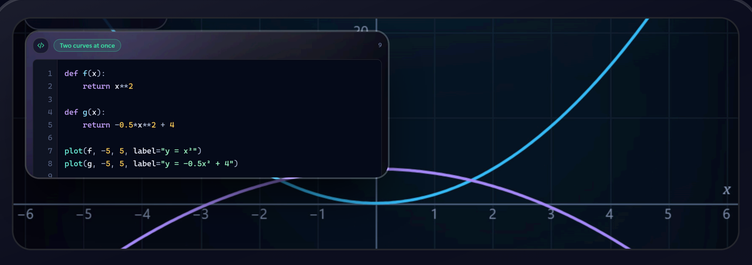
03 INSIDE THE LAB — Real screens, real mathematics

Every image is captured from the working system — nothing is mocked up.



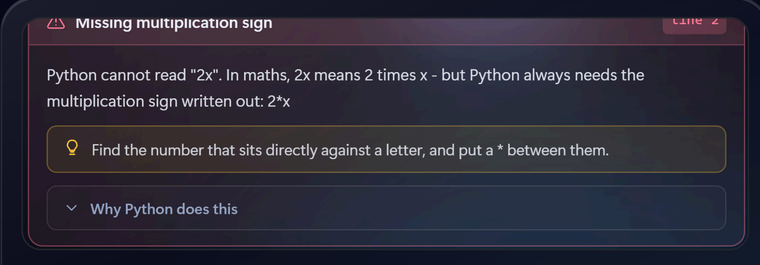
What If? — predict, then see

Students commit to a prediction before anything moves. Then the curve animates against a ghost of the old one, and the lab explains why it changed.



Every curve is also a program

Write Python, press Run, get mathematics back: **code → data → graph**, drawn on as you watch.



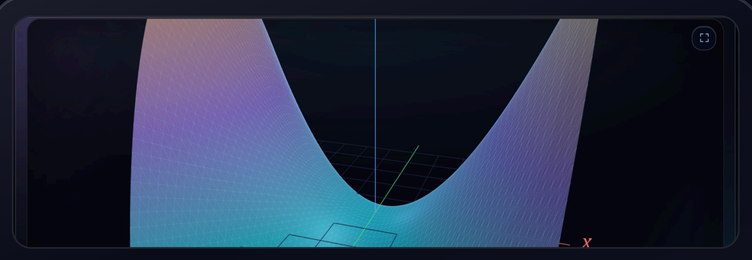
Mistakes become lessons

Type `y = 2x + 3` and the lab explains that maths allows $2x$ but Python needs `2*x` — with a hint that stops short of the answer.



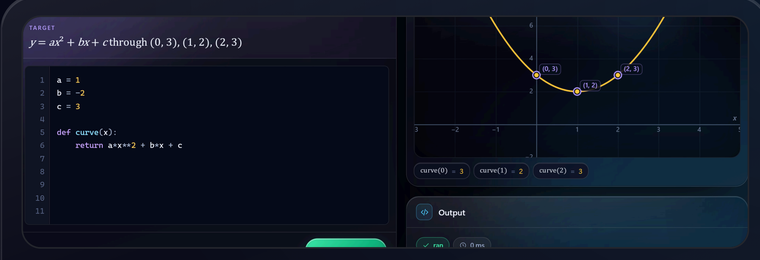
Shapes that know their maths

`from shapes import *` — polygons, solids and turtle graphics. **Every shape reports its area, perimeter and volume** live.



Maths in three dimensions

Rotate $z = a(x^2 - y^2)$, then slice it: **every cross-section is an ordinary 2D curve**, drawn alongside.



Graded by running the code

Answers are run, not text-matched, so **any correct solution passes**. The student's function is drawn over the goal.

04 WHAT MAKES IT NEW

● yes ◐ partly / some products ○ no — n/a

	GRAPHING APPS	ONLINE PYTHON EDITORS	MATHXPLORE
Change maths live, in 2D and 3D	●	○	●
Python tied to the maths on screen	○	◐	●
Predict-then-reveal, built in	◐	○	●
Errors explained in maths terms	—	○	●
Offline · no account · no data sent	◐	○	●

05 IMPACT

Students
Understanding over memorising: every change is predicted, seen and explained.

Teachers
Ready demonstrations, plus a class view built from students' real exported records.

Low-connectivity schools
Runs from one laptop or a school intranet — no internet needed after setup.

Evidence-ready
No effect claimed yet — anonymous on-device logs are ready for a pre/post-test study.

06 MARKET & NEXT STEPS

WHO	Upper-secondary & pre-university students, teachers, tutors
MODEL	Free for students · school licence · offline school kit
COST	Static files, no server — near-zero cost per extra student
NEXT	School pilot (pre/post-test) · Bahasa Melayu UI · authoring

07 UNDER THE HOOD

`your Python` → `parser` → `interpreter` → `graphs · shapes · lessons`

- ✓ Own Python engine, ~3,900 lines, no dependencies
- ✓ No `eval`, no network, no file access
- ✓ Runaway loops stopped safely
- ✓ React · TypeScript · Three.js

QR code demo link or video

Try it yourself
Ask us for the keyboard, or scan to open the lab.

your.email@domain